

Full length article

Latent diffusion-driven inverse design of damping microstructures with multi-axial nonlinear mechanical targets

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ABSTRACT

This study presents an integrated generative framework for the inverse design of damping microstructures in energy-dissipating steel walls (EDSWs) for seismic applications, establishing a seamless pipeline from large-scale pixel-based datasets to latent-space representation, three-dimensional reconstruction, industrial fabrication, and finite element analysis (FEA) verification. Starting from over 140,000 boundary-identical microstructures, a variational autoencoder-based TopoFormer compresses geometric features into latent codes, enabling over 90% reduction in generation complexity while maintaining high reconstruction fidelity. Representative structures are selected via k-means clustering in the latent space and analyzed through nonlinear FEA under shear and compression to construct a performance-labeled dataset. A conditional latent diffusion transformer (DiT) is then trained to map complete nonlinear mechanical performance curves to manufacturable geometries, thus achieving a one-to-many correspondence between target responses and structural configurations. Comparative evaluations show that the proposed DiT framework surpasses multiple CondUNet baselines, achieving the lowest FID (11.367) and the highest SSIM (0.676) with balanced coverage and precision. Experimental validation using laser-cut low-yield-point steel specimens under low-cycle reciprocating loading demonstrates close agreement between generated and target hysteresis curves, confirming both geometric fidelity and mechanical reliability. The results establish a scalable, high-accuracy, and experimentally validated approach for automated, performance-driven microstructure design, providing a practical pathway for incorporating generative artificial intelligence into the engineering development of next-generation seismic energy-dissipation systems. The related codes are available at <https://github.com/AshenOneme/DiT-Based-Microstructures-Design>.

1. Introduction

Earthquakes rank among the most devastating natural hazards, often resulting in extensive structural damage and profound social and economic disruption. Field investigations conducted after major seismic events have consistently revealed that insufficient seismic performance of buildings remains a principal cause of structural collapse, significant casualties, and substantial economic losses [1–6]. To mitigate the destructive impacts of earthquakes, various seismic design methodologies have been developed. Compared with conventional seismic design, which primarily relies on enhancing the inherent strength and stiffness of structures to passively resist seismic energy, energy-dissipation and vibration-control technologies provide significant and unparalleled advantages in terms of both safety and cost-effectiveness. Based on their underlying energy-dissipation mechanisms, dampers can generally be classified into displacement-dependent and velocity-dependent types.

Typical displacement-dependent devices include buckling-restrained braces [7,8], metallic yielding dampers [9,10], friction dampers [11,12], whereas velocity-dependent devices are usually composed of viscoelastic [13,14] or viscous materials, with representative applications such as viscous dampers [15,16] and viscous damping walls [17,18]. Although the aforementioned energy-dissipating measures can enhance the seismic performance of building structures to some extent, the current practice of adopting concentrated dampers presents several limitations. Such systems often occupy considerable space and conflict with the placement of non-structural infill walls. Moreover, concentrated energy dissipation may induce local stiffness irregularities in the floor plan, thereby increasing the risk of torsional responses caused by eccentricity between the centers of mass and rigidity.

To address the limitations of conventional energy-dissipation approaches, Zhang et al. [19] proposed a distributed energy-dissipation method that integrates an energy-dissipating steel wall (EDSW) with

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infill walls, as illustrated in Fig. 1. This novel system not only enhances the overall seismic performance of building structures but also improves the seismic resilience of non-structural infill walls, which are often neglected in seismic design. An EDSW consists of an array of one or multiple types of damping microstructures. Since its thickness is constrained by the presence of infill walls, the design is generally limited to the in-plane configuration to satisfy energy-dissipation demands at different locations. Under seismic loading, the EDSW dissipates energy as a whole; however, the deformation modes of individual damping microstructures vary significantly, leading to considerable differences in their mechanical performance. While the overall mechanical behavior of the EDSW can be approximately evaluated by decomposing its constituent microstructures, the mechanical design of each individual microstructure remains challenging due to their structural heterogeneity. At present, most studies on damping microstructures focus on experimental investigations of their impact or compressive performance, which are generally characterized in the metamaterials field as energy absorption [20–22]. In contrast, seismic engineering emphasizes energy dissipation through hysteresis behavior rather than unidirectional compression. The design of damping microstructures still relies heavily on experimental approaches. Although traditional lattice structures have been widely applied in impact energy-absorption studies, their applicability in seismic engineering is limited because the low-cycle fatigue performance required for EDSWs is not fully satisfied by such metamaterial-based unit cells [23].

With the rapid development of artificial intelligence, particularly the emergence of generative AI [24–27], significant progress has been achieved in the inverse design of energy-dissipating microstructures [28–31]. By leveraging data-driven learning and optimization, generative AI provides powerful tools for exploring complex design spaces and offers new avenues for automated and efficient structural innovation [32–36]. The aforementioned studies established large-scale microstructural datasets [37–39] and employed finite element methods to investigate the mechanical properties of the corresponding microstructures, thereby promoting the engineering application of damping microstructures. However, most existing approaches primarily focus on directional stiffness and equivalent Poisson's ratio. Although such methods considerably broaden potential application scenarios, they are not well suited to strongly nonlinear problems, such as the seismic excitation of EDSWs. Therefore, dedicated research is required to develop damping microstructure datasets tailored for seismic applications and to establish inverse design methods guided by mechanical performance targets.

While existing inverse design frameworks for microstructures have achieved considerable progress, the method proposed in this study further advances the field in the following aspects:

- A latent Diffusion Transformer based inverse design model is employed, in which a self-supervised discriminator is incorporated to supervise the generation process. This strategy not only enhances generation accuracy but also reduces computational complexity, thereby achieving a balance between efficiency and accuracy.
- The proposed framework explicitly accounts for the multidirectional nonlinear behavior of damping microstructures. Instead of using scalar mechanical indicators as inputs, the model accepts complete performance curves, thereby enabling a one-to-many mapping from one-dimensional mechanical response to two-dimensional structural configurations.
- A batch-oriented modeling and finite element analysis (FEA) strategy is developed, establishing an integrated pipeline from pixel space to three-dimensional models, and further from physical fabrication to numerical simulation.

2. Methodology

2.1. Overview of the workflow

The microstructure generation framework proposed in this study is based on the boundary-identical microstructures dataset introduced by Feng et al. [28]. This dataset exhibits a sufficiently broad stiffness distribution and strong practical applicability. However, the original dataset only provides a large number of 256×256 -pixel geometric patterns with their corresponding effective elastic module, without finite element modeling or nonlinear analysis. In this work, more than 140,000 distinct microstructures from the original dataset are employed, and the overall workflow is presented in Fig. 2.

Direct training or generation based on the original dataset involves high computational complexity. To mitigate this, a TopoFormer model based on a variational autoencoder (VAE) is first trained to transform the dataset from the pixel space into a latent space. Subsequent microstructure generation is then carried out entirely within the latent space, as illustrated in Fig. 2(b). The original dataset contains nearly 150,000 different patterns; however, many of them exhibit similar structural features. Considering that nonlinear FEA entails substantial computational cost, k-means clustering was performed in the latent space to select approximately 10,000 representative patterns, as shown in Fig. 2(c). These selected patterns were then subjected to finite element modeling and nonlinear analyses under compression and shear loading, as illustrated in Fig. 3(d). The mapping between target mechanical properties and the latent space is learned through a diffusion model, after which the latent representations are decoded back into the pixel space using a VAE decoder, as illustrated in Fig. 2(f). Previous research has shown that imposing physical constraints during the generation

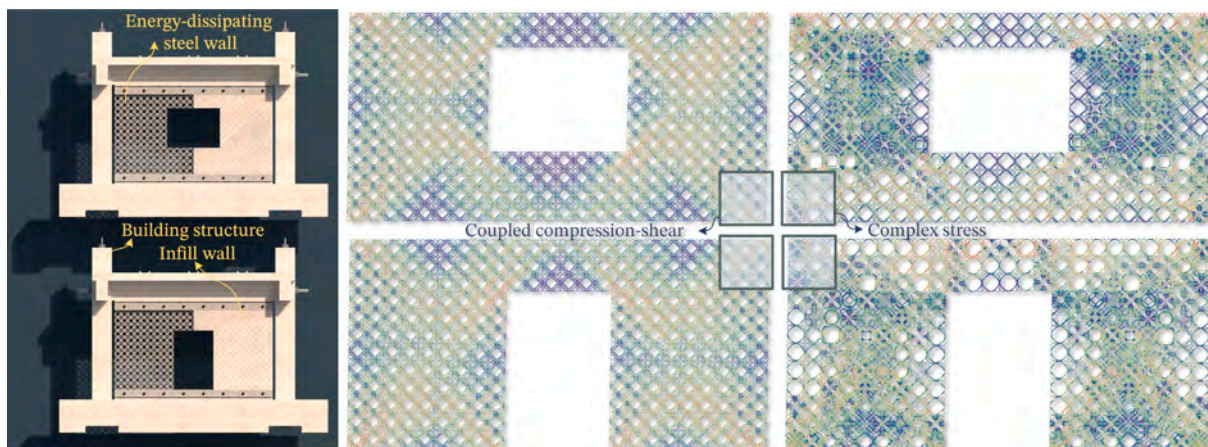


Fig. 1. Seismic deformation characteristics of EDSWs.

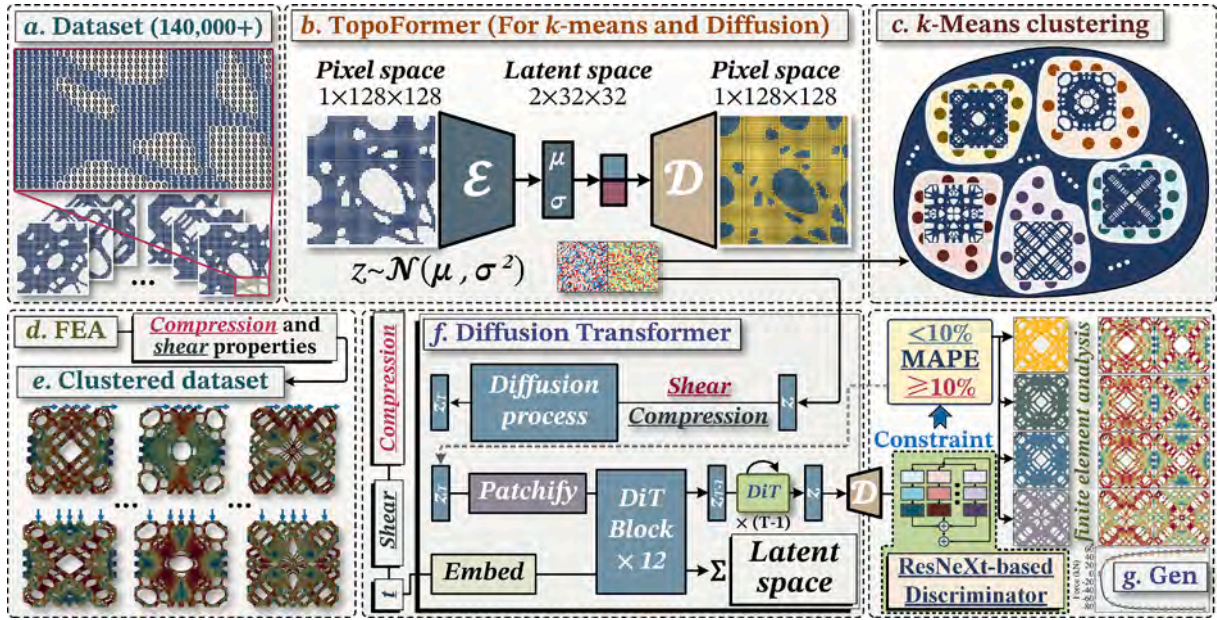


Fig. 2. Workflow of the generative framework for damping microstructures.

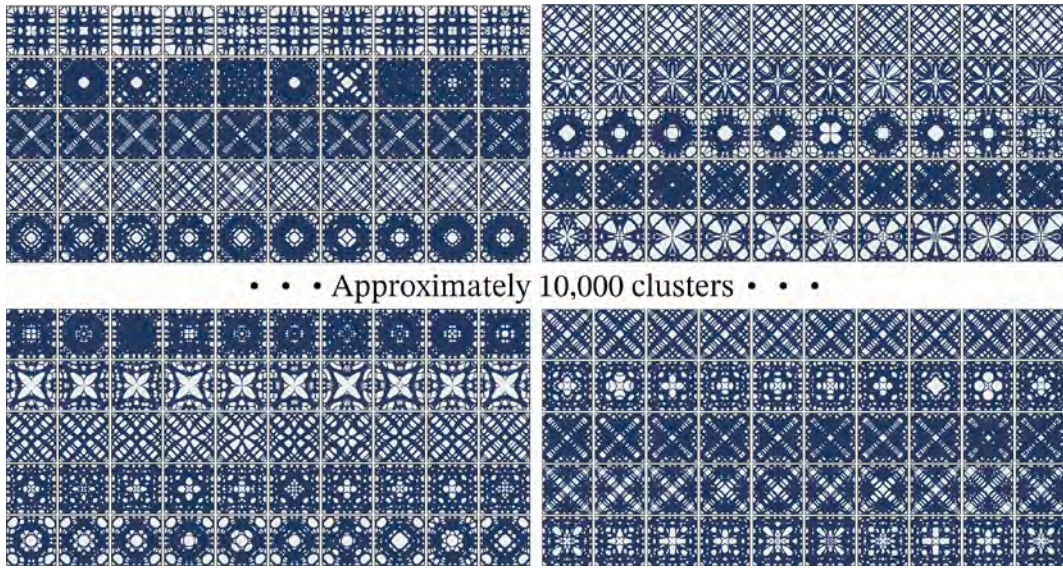


Fig. 3. Clustering results.

process enables the synthesized microstructures to better conform to the desired properties. Considering that the fine-scale features of microstructures exert a significant influence on their overall performance, and that ResNeXt [40] is particularly effective in capturing such features, this study employs ResNeXt as a pretrained mechanical property discriminator. The final configurations that satisfy the discrimination criteria are then subjected to FEA to verify whether their performance approaches the target values, as shown in Fig. 2(g).

2.2. Dataset simplification

The original dataset contains 141,636 distinct microstructures, which, despite being different, share many redundant features. Processing such a large dataset poses substantial challenges for finite element analysis and algorithmic development, as building complete models for all microstructures would greatly increase computational cost and is unnecessary for generative tasks. To address this, we first

extracted latent representations using the TopoFormer model, effectively simplifying complex features, facilitating diffusion model training, and enabling clustering in a reduced-dimensional space.

Clustering was conducted in the TopoFormer-encoded latent space of size $2 \times 32 \times 32$, corresponding to a 96.88 % reduction relative to the pixel domain, while preserving structured, geometry-aware features. Latent vectors were normalized, and silhouette coefficient analysis yielded an average of 0.642. t-SNE visualization confirmed clear separability among groups. On a subset of 20,000 samples, k-means achieved clustering quality comparable to agglomerative clustering and DBSCAN, but required approximately 9.9 and 21.6 times less computation time, respectively. The final selection of 10,000 cluster centers preserved more than 97.65 % of the variance in shear and compressive strength distributions, ensuring representativeness for subsequent finite element analysis and generative model training.

The selection of k is 10,000 was based on empirical evaluation of variance preservation in mechanical property distributions and

consideration of FEA cost. This value maintains over 97.65 % of property variance while keeping the total number of required simulations within the available computational budget, ensuring both representativeness and feasibility. A portion of the clustering results is presented in Fig. 3. It can be observed that the ten microstructures in each row exhibit highly similar features. Therefore, one microstructure was randomly selected from each group to construct the new dataset. Following this approach, approximately 10,000 clusters were extracted from the original dataset of more than 100,000 microstructures.

2.3. Mechanical properties of microstructures

In the numerical simulations, low-yield-point steel with a nominal yield strength of 235 MPa, commonly used for metallic dampers, was adopted as the material. A total of 10,052 finite element models of microstructures with dimensions of 100×100 mm were analyzed under shear and compressive loading, and the results are presented in Fig. 4. Here, the positive direction of force corresponds to shear resistance, while the negative direction corresponds to compressive resistance. The deformation of the microstructures is characterized by drift. The force–drift curves indicate that, for most energy-dissipating microstructures, the compressive resistance is significantly higher than the shear resistance. For EDSWs, the microstructures in different regions are subjected to distinctly different shear and compressive deformations. Therefore, by combining microstructures with diverse mechanical properties, the overall stress distribution of EDSWs can be effectively tuned, which further highlights the significance of this study. Representative microstructures with varying mechanical properties are shown in Fig. 4(a)–(j), arranged approximately in order of increasing load capacity. It is evident that their volume fraction gradually increases, and the microstructure with the maximum load capacity contains almost no internal voids, except for boundary cutouts designed to facilitate the connection between adjacent microstructures, as their boundaries remain identical.

The mechanical performance distributions of different microstructures are shown in Fig. 5, derived from their shear and compressive response curves. Overall, the shear and compressive energy dissipation capacities exhibit a positive correlation, but with considerable scatter. For example, some microstructures with high shear resistance show relatively low compressive resistance, as illustrated in Fig. 5(a). From the perspective of volume fraction, the yielding capacities in different loading directions also show weak correlations. Across the entire dataset, the maximum shear yield strength reaches 72.77 kN, while the minimum is only 0.48 kN; the maximum compressive resistance is 249.07 kN, whereas the minimum is 1.83 kN. These results indicate that the mechanical properties of the dataset are highly diverse and scattered, which poses significant challenges for inverse design algorithms.

2.4. Conditional diffusion models

Conditional Diffusion Models (CDMs) are a class of deep generative models built upon probabilistic diffusion processes [41–45]. The core

idea is to progressively inject noise into the data in the forward process until it approaches isotropic Gaussian noise. During the reverse process, a neural network, guided by the generative conditions, learns to denoise step by step, thereby recovering the original data distribution and enabling the generation of high-quality samples from random noise.

In the forward diffusion process, an original sample $x_0 \sim q(x_0)$ is gradually perturbed by Gaussian noise through a Markov chain with the transition distribution:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I) \quad (1)$$

where $\{\beta_t\}_{t=1}^T$ denotes a predefined noise schedule. For notational simplicity, we define $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. Accordingly, the marginal distribution at step t can be written as:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I) \quad (2)$$

from which direct sampling is available in closed form:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \epsilon \in \mathcal{N}(0, I) \quad (3)$$

To construct the reverse generative process, the posterior distribution of the forward process is required. Based on the properties of Gaussian chains, the posterior can be expressed as:

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I) \quad (4)$$

with mean and variance given by:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t x_0 + \sqrt{\bar{\alpha}_t} (1 - \bar{\alpha}_{t-1}) x_t}{1 - \bar{\alpha}_t} \quad (5)$$

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t \quad (6)$$

The reverse diffusion process is parameterized by a learnable neural network, in which the conditional variables are explicitly introduced as the shear target (s) and the compression target (c). The transition distribution is defined as:

$$p_\theta(x_{t-1}|x_t, s, c) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, \text{TimeEmb}(t), s, c), \sigma_\theta^2 I) \quad (7)$$

where the mean is estimated by the noise prediction network $\epsilon_\theta(x_t, \text{TimeEmb}(t), s, c)$:

$$\mu_\theta(x_t, t, s, c) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, \text{TimeEmb}(t), s, c) \right) \quad (8)$$

During training, the model is optimized using a noise matching loss that minimizes the mean squared error between the predicted noise and the true noise:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} \left[\left\| \epsilon - \epsilon_\theta(x_t, \text{TimeEmb}(t), s, c) \right\|_2^2 \right] \quad (9)$$

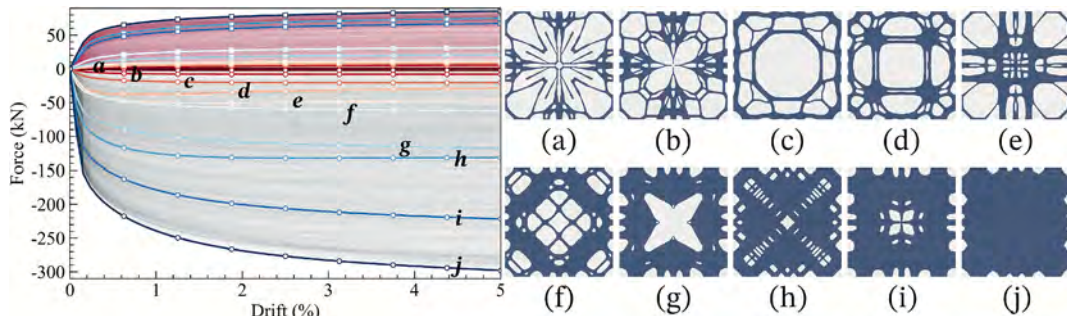


Fig. 4. Force–drift curves under shear and compressive loading.

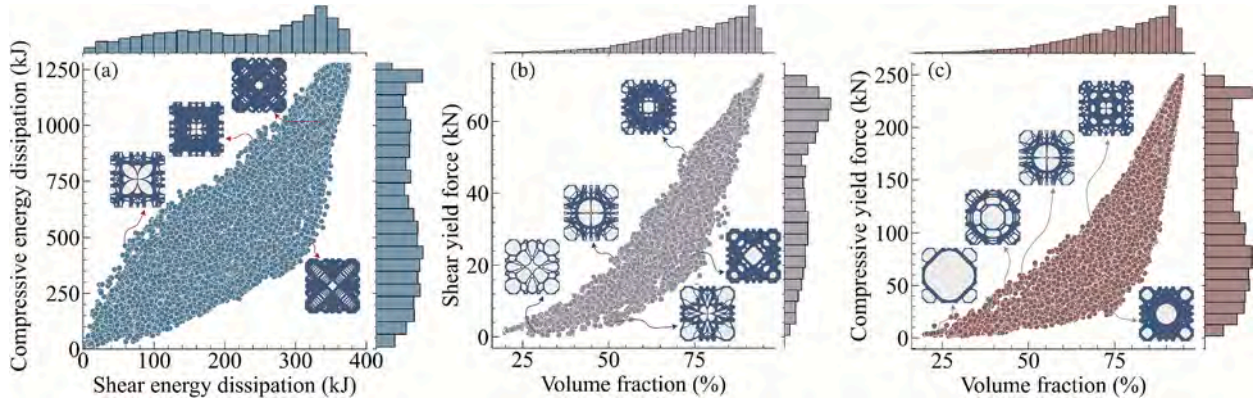


Fig. 5. Distribution of mechanical properties.

to further improve the quality of conditional generation, classifier-free guidance [46] is employed. Specifically, both conditional and unconditional noise predictors are jointly trained, and their outputs are linearly combined during sampling to obtain the guided noise prediction:

$$\hat{\epsilon}_{\theta}(x_t, \text{TimeEmb}(t), s, c) = (1 + \omega)\epsilon_{\theta}(x_t, \text{TimeEmb}(t), s, c) - \omega\epsilon_{\theta}(x_t, \text{TimeEmb}(t), \emptyset) \quad (10)$$

where the hyperparameter ω controls the strength of conditional guidance, thereby balancing fidelity and diversity in the generated microstructures.

2.5. Architectural design

In conditional generation tasks of diffusion models, the design of the network architecture plays a decisive role in both model performance and generation quality. The most widely adopted architectures include the conditional U-Net (CondUNet) and the Transformer-based Diffusion Transformer (DiT), which represent typical implementations of convolutional neural networks and Transformer frameworks, respectively, in diffusion modeling.

CondUNet is an extension of the U-Net architecture with the explicit incorporation of conditional variables. The original U-Net employs a symmetric encoder-decoder structure with skip connections, thus effectively preserving multi-scale features. Building on this foundation, CondUNet injects conditional information through multiple mechanisms: conditional embedding vectors are incorporated into the feature maps, and conditional signals are further integrated within the attention layers. This design enables CondUNet to retain strong local spatial modeling capabilities while flexibly enforcing consistency between the generated outputs and external conditions (e.g., shear target ss and compression target), as illustrated in Fig. 6. In this study, by varying the number of layers in the ResBlock2D and AttnBlock modules [47], we construct four CondUNet variants, namely CondUNet-S, CondUNet-B, CondUNet-L, and CondUNet-X. These suffixes follow the common scaling convention in deep learning architectures, representing Small, Base, Large, and Extra-Large configurations, similar to the naming strategies adopted in Vision Transformers [48,49], where model size increases with depth and width.

DiT represents a more recent line of research that replaces the U-Net backbone with a pure Transformer architecture. In DiT, the intermediate state x_t of the diffusion process is partitioned into patches or serialized as

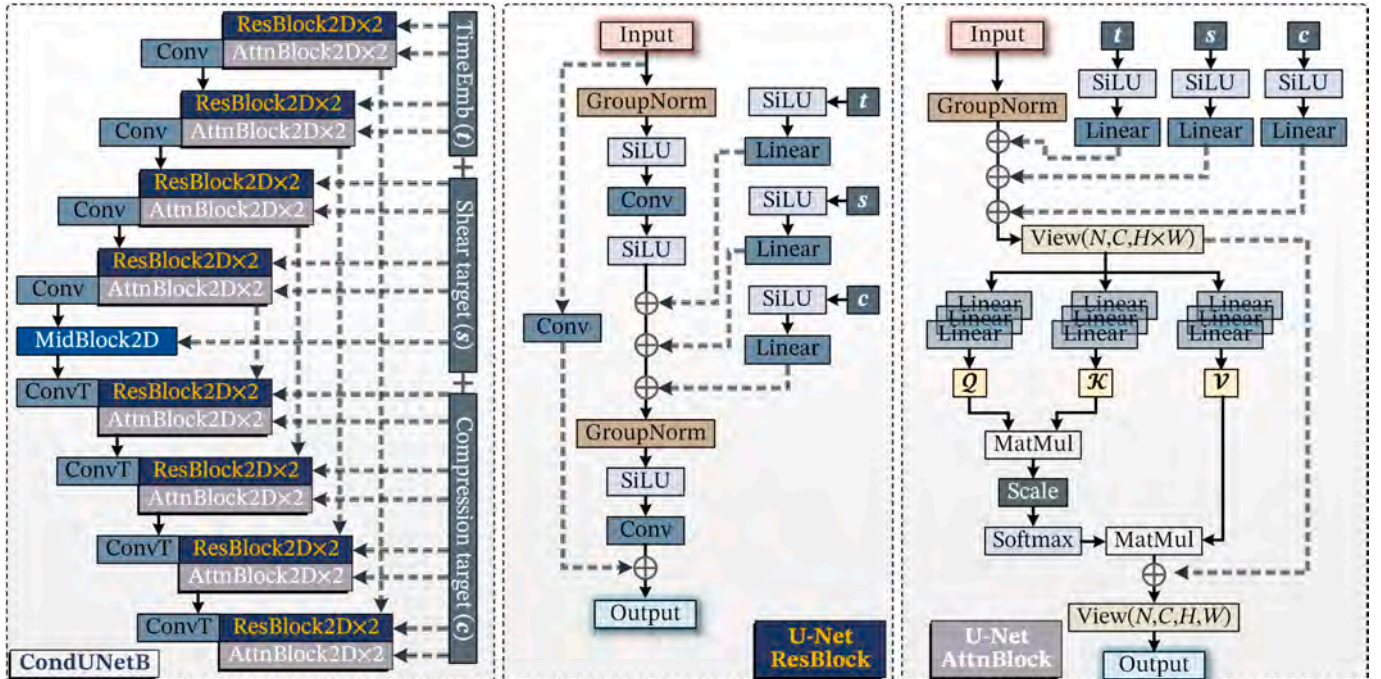


Fig. 6. Architecture of the CondUNet network.

feature vectors, which are then modeled using standard Transformer blocks. Its primary advantage lies in the self-attention mechanism, which effectively captures long-range dependencies. Furthermore, conditional control is achieved via cross-attention with conditional embeddings or through token-level fusion, as shown in Fig. 7. As a complementary approach, we also establish and evaluate DiT-based models in this study, enabling a direct comparison with CondUNet, as shown in Fig. 8.

3. Finite element analysis results and experiments

3.1. Specimen preparation

From the clustering results shown in Fig. 3, six clusters were randomly selected for specimen fabrication. Using the TXT2CAE plugin for Abaqus, developed in our previous work, available at <https://github.com/AshenOneme/MultiDampGen>, a large number of pixel-space images can be automatically converted into three-dimensional models. During this process, the generated models can be directly meshed for subsequent FEA, or exported as 3D geometries for specimen fabrication.

A rapid, convenient, and cost-effective fabrication approach based on laser cutting was employed. Once the 3D model was obtained, the corresponding patterns could be directly cut with a dimensional accuracy of ± 0.1 mm. The cutting process is generally very fast, and even complex microstructures with numerous perforations can be completed within one minute. Subsequently, a shot blasting process was applied to remove residual molten slags and to clean the specimen surface, as illustrated in Fig. 9.

3.2. Experimental verification

To better simulate the deformation behavior of the microstructure under seismic loading, low-cycle reversed shear loading tests were conducted to validate the reliability of the finite element modeling developed in the preliminary stage. The experimental setup is shown in Fig. 10. For ease of installation, connecting components were added to the top and bottom of the microstructure. Bolted joints were adopted both between the microstructure and the connectors, and between the connectors and the actuator. In addition, displacement transducers were employed to record the actual drift of the damping microstructure.

The comparison between the failure patterns of the damping microstructure obtained from the finite element model and those observed in the experiments is shown in Fig. 11. The experimental results provide strong validation of the reliability of the finite element simulations, as the predicted failure locations and modes closely match the experimental observations. The hysteresis responses of the six specimens are compared in Fig. 12. The FEA derived curves exhibit excellent agreement with the experimental counterparts. Except for slight slips caused by imperfect closure of the testing device at higher load levels, the hysteresis loops from both approaches show highly consistent features in terms of shape, stiffness, energy dissipation, and yield points. In all finite element simulations, the damping microstructures are modeled using low-yield-point steel, which is widely adopted in metallic damper applications. The constitutive law incorporates combined isotropic-kinematic hardening to accurately capture cyclic loading behavior. The detailed material parameters used in ABAQUS are summarized in Table 1.

The experimental results confirmed the reliability of the proposed modeling approach. However, when the drift exceeded 5 %, each model required a considerable amount of computational time. To simplify the analysis, the dataset in this study was established with loading applied up to 5 %. For seismic loading scenarios, this level of deformation is already considered significant.

4. Implementation details

4.1. Evaluation method

To quantitatively evaluate the quality of the generated damping microstructures with respect to the reference ground truth, we adopt four widely used generative evaluation metrics: the Fréchet Inception Distance (FID), Coverage (COV), Precision (PREC), and Structural Similarity Index (SSIM). These indicators jointly capture distributional consistency, sample diversity, fidelity, and structural similarity, thereby providing a comprehensive assessment of the proposed generation method.

The FID is employed to measure the distributional discrepancy between the generated microstructures and the reference set. Assuming that the feature distribution of real (ground truth) microstructures follows $\mathcal{N}(\mu_r, \Sigma_r)$ and that of generated microstructures follows $\mathcal{N}(\mu_g, \Sigma_g)$,

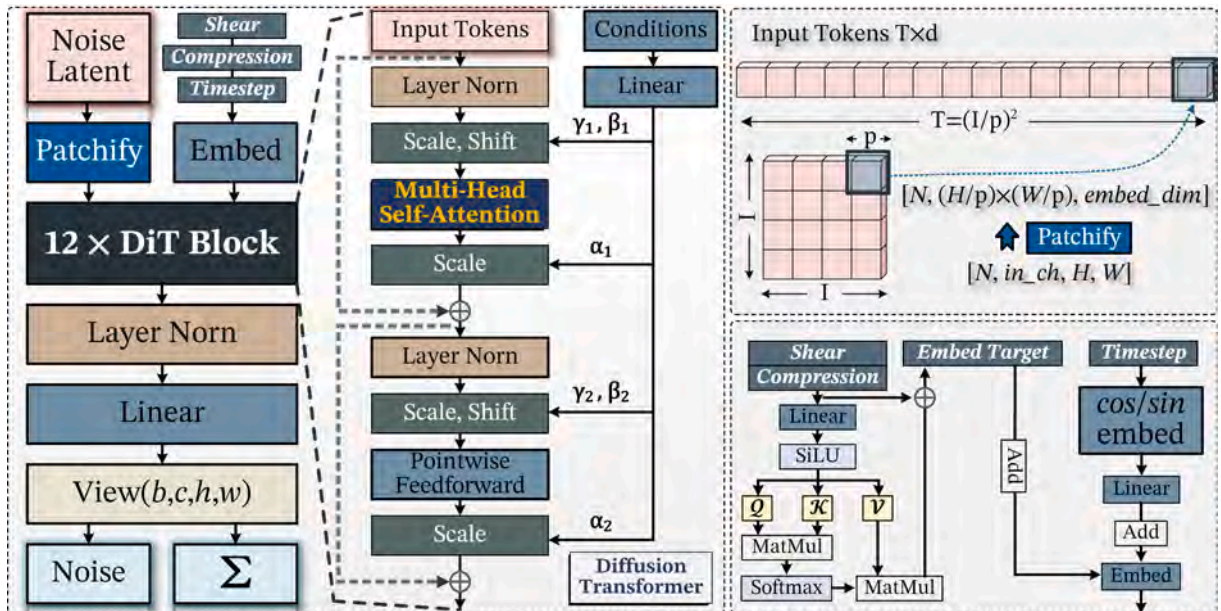


Fig. 7. Architecture of the Diffusion Transformer network.

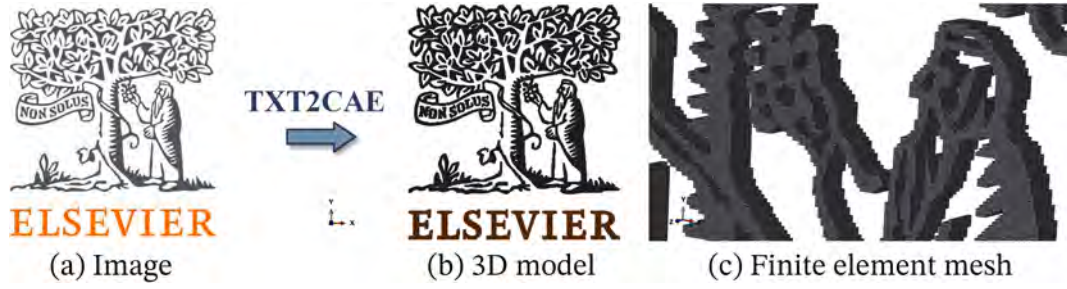


Fig. 8. TXT2CAE tool.

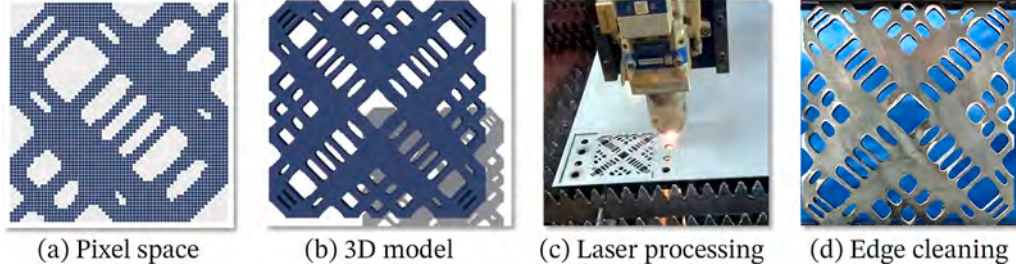


Fig. 9. Specimen fabrication process.

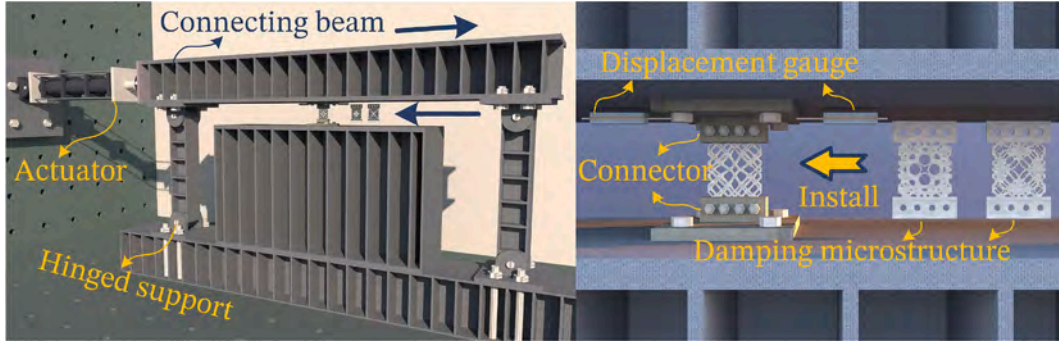


Fig. 10. Loading system and specimen installation.

the FID score is defined as:

$$FID = \|\mu_r - \mu_g\|_2^2 + Tr(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}) \quad (11)$$

A lower FID indicates that the generated distribution is closer to the reference distribution in terms of feature space, reflecting better generative quality.

To further assess the balance between fidelity and diversity, COV and PREC are adopted. Formally, let $\{x_i^r\}_{i=1}^N$ denote the set of ground truth microstructures and $\{x_j^g\}_{j=1}^{N_g}$ denote the set of generated microstructures. For each sample, we compute its nearest neighbor distance in the feature space. COV measures the fraction of real samples that are covered by the generated distribution, defined as:

$$COV = \frac{1}{N_r} \sum_{i=1}^{N_r} 1\left(\exists x_j^g \text{ s.t. } d(x_i^r, x_j^g) \leq \min_{k \neq i} d(x_i^r, x_k^r)\right) \quad (12)$$

where $d(\bullet, \bullet)$ denotes the distance metric in feature space, and $1(\bullet)$ is the indicator function. A higher COV indicates broader coverage of the real data distribution, thus reflecting greater diversity of generated microstructures. PREC evaluates the proportion of generated samples that fall within the manifold of real microstructures, defined as:

$$PREC = \frac{1}{N_g} \sum_{j=1}^{N_g} 1\left(\exists x_i^r \text{ s.t. } d(x_j^g, x_i^r) \leq \min_{k \neq i} d(x_i^r, x_k^r)\right) \quad (13)$$

A higher PREC indicates that generated samples are closer to the real manifold, reflecting higher fidelity and structural plausibility. Together, COV and PREC provide complementary insights into the trade-off between exploring diverse structural patterns and ensuring that the generated results remain realistic.

In addition to distributional measures, the SSIM is employed to directly compare the geometry and morphology of individual generated microstructures with their ground truth counterparts and is defined as:

$$SSIM(x, y) = \frac{(2\mu_x \mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (14)$$

where μ_x, μ_y are the mean values, σ_x^2, σ_y^2 are the variances, σ_{xy} is the covariance, and C_1, C_2 are small constants to stabilize the computation. A higher SSIM score indicates a closer structural match between generated and reference microstructures.

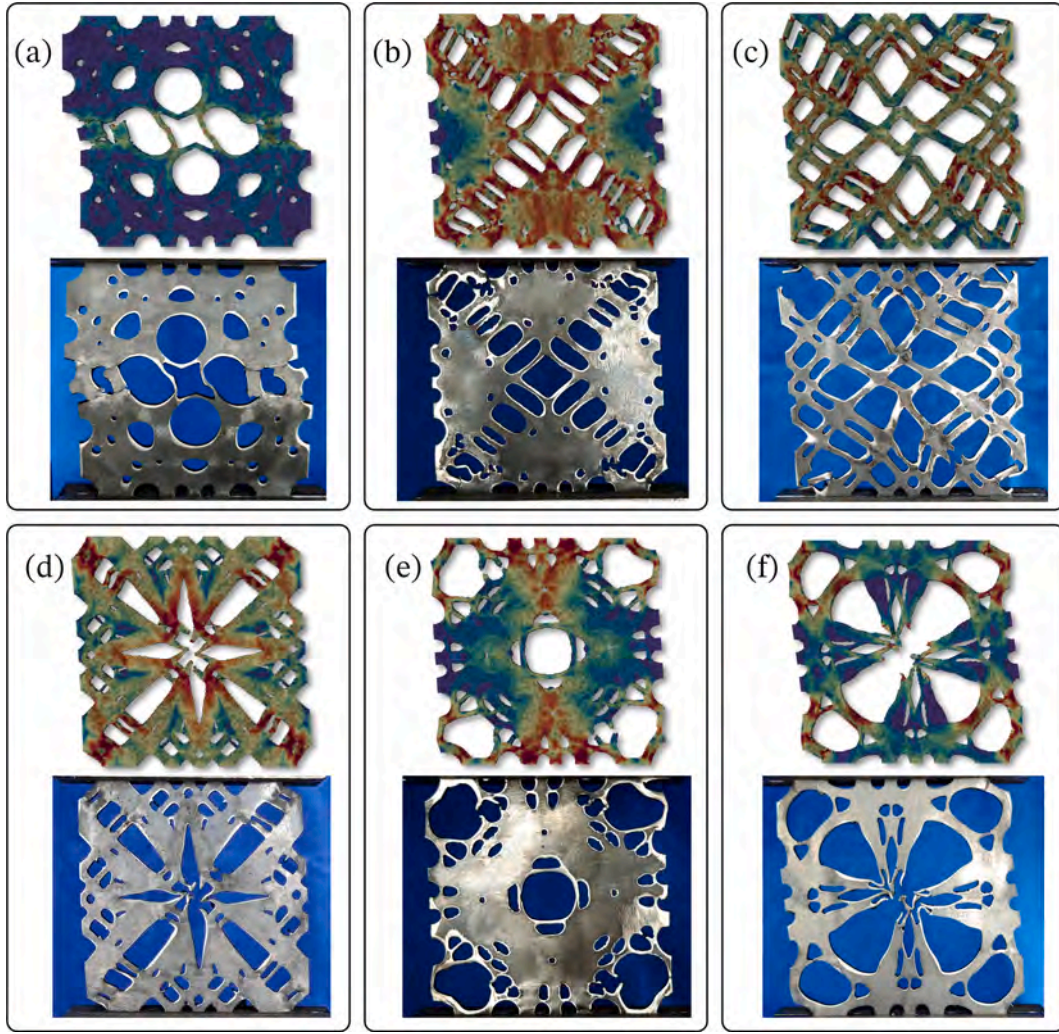


Fig. 11. Loading system and specimen installation.

4.2. Experimental setup

The experimental work in this study was carried out on a Windows 10 Professional platform. For dataset generation, finite element simulations were conducted with ABAQUS 2023, while model training was implemented in a PyTorch framework supported by dual NVIDIA RTX 3090 GPUs, as summarized in Table 2.

4.3. Parameter details

The training of the microstructure generation framework was optimized using the AdamW algorithm with an initial learning rate of 1×10^{-4} , following a cosine learning rate schedule with linear warm-up. A weight decay of 0.01 was applied to stabilize optimization. The model was trained for up to 1,000 iterations with a batch size of 32, distributed across multiple GPUs. For the diffusion process, the noise scheduler followed the standard DDPM [46] formulation, with the total number of time steps set to 1000. The variance schedule β_t increased linearly from 1×10^{-4} to 0.02. During inference, a DDIM sampler [41] with 200 timesteps was employed to accelerate generation while preserving sample quality. Normalization and regularization techniques were also adopted to enhance training stability. Specifically, layer normalization was applied within the DiT blocks, and gradient clipping with a maximum norm of 1.0 was enforced. A detailed summary of the hyperparameter settings is provided in Table 3.

5. Results and discussion

5.1. ResNeXt-based discriminator

As a key component imposing physical constraints during the generation process, the prediction accuracy of the ResNeXt-based discriminator directly determines the quality of the generated samples. This quality is not merely reflected in the geometric continuity and completeness of the patterns, but more critically in whether the generated results can meet the predefined performance requirements. In this study, four backbone architectures, namely ResNet18, ResNeXt18, ResNeXt50, and ResNeXt101, were employed for discriminator training. The results show that the final MSE loss of ResNeXt18 was reduced by 99.64 % compared with ResNet18, while ResNeXt101 yielded only a further 34.59 % reduction relative to ResNeXt18. These findings demonstrate that ResNeXt can achieve high predictive accuracy for microstructural performance with relatively low parameter complexity. Fig. 13 presents the comparison between the actual mechanical properties and the predicted values for representative microstructures. It is observed that for microstructures with higher load-bearing capacity, the ResNeXt-based discriminator provides more accurate predictions. However, for cases with very low compressive capacity, certain deviations occur in the predictions of compressive strength, whereas the predictions of shear performance remain highly consistent with the FEA results.

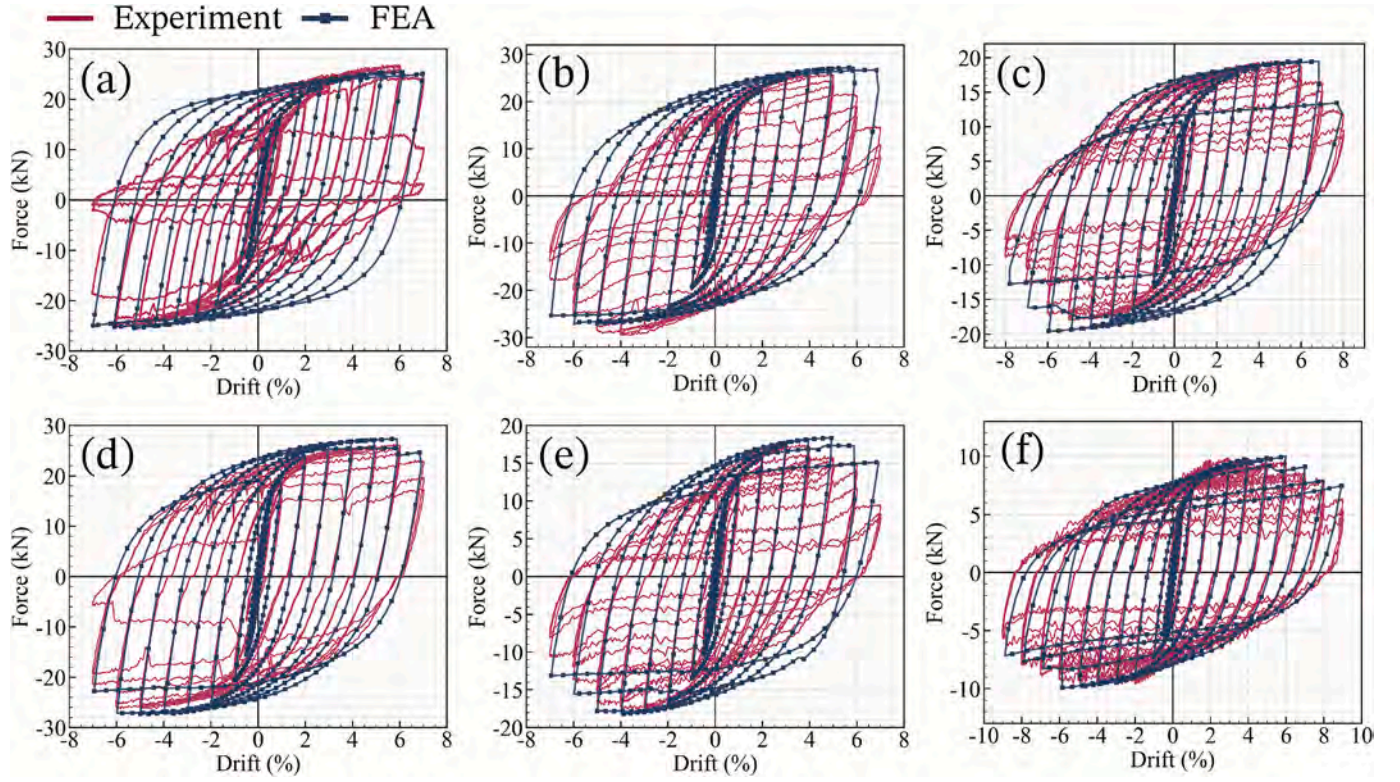


Fig. 12. Loading system and specimen installation.

Table 1

Material model parameters for FEM simulations.

σ_y	E_y	Q_{∞}	b	C_1	γ_1	C_2	γ_2	C_3	γ_3	C_4	γ_4
275.17	206,000	40	1.2	30,000	1000	15,000	100	600	20	350	10

Table 2

Environment configuration.

Category	Component	Specification
Hardware	CPU	AMD EPYC 9654 96-Core Processor
	GPU	2 × NVIDIA 3090 GPU
	RAM	480 GB
Software	System	Windows 10 Professional version
	Python	3.12
	ABAQUS	2020
Environment	PyTorch	2.6
	CUDA	12.4

Table 3

Parameter settings.

Category	Parameter	Value
AdamW optimizer	Learning rate	0.0001
	Weight decay	0.01
Lr scheduler	Cosine schedule + warmup	100
Noise scheduler (train)	Timesteps	1000
	β_{schedule}	Squared cosine
	β_{start}	0.0001
	β_{end}	0.02
Sampling strategy (inference)	Timesteps	200
	Guidance scale	4.0

5.2. Results and validation

The five trained generative models were applied to the same shear and compressive targets, and each generated four structural configurations for FEA. Fig. 14 presents the comparison between the mechanical properties of these microstructures and the corresponding predefined mechanical targets. The different structures generated by the five models are generally close to the predefined targets, and many microstructures exhibit considerable similarity in morphology. However, even when the geometrical appearances are nearly identical, noticeable differences may arise in their mechanical performance. For example, the last microstructure generated by CondUNet-B and CondUNet-L shows an extremely high visual resemblance, yet their compressive strengths differ by 10.77 %.

The microstructure with the highest generation frequency was selected for fabrication to further verify the reliability of both the diffusion model and the finite element simulations. The low-cycle reciprocating loading process is illustrated in Fig. 15, where the positive drift of 10 % is shown in Fig. 15(a) and the negative drift of 10 % in Fig. 15(b). Under these two displacement conditions, the fabricated microstructure exhibited severe damage, leading to a sharp reduction in load-bearing capacity, which was accurately captured by the finite element simulations. It is worth noting that no pronounced out-of-plane buckling was observed in the generated microstructures during the cyclic loading process, and the ultimate drift reached 10 %. Moreover, the specimens exhibited a progressive failure pattern at later stages. These results demonstrate that the proposed generative model holds substantial potential for discovering microstructures with superior mechanical

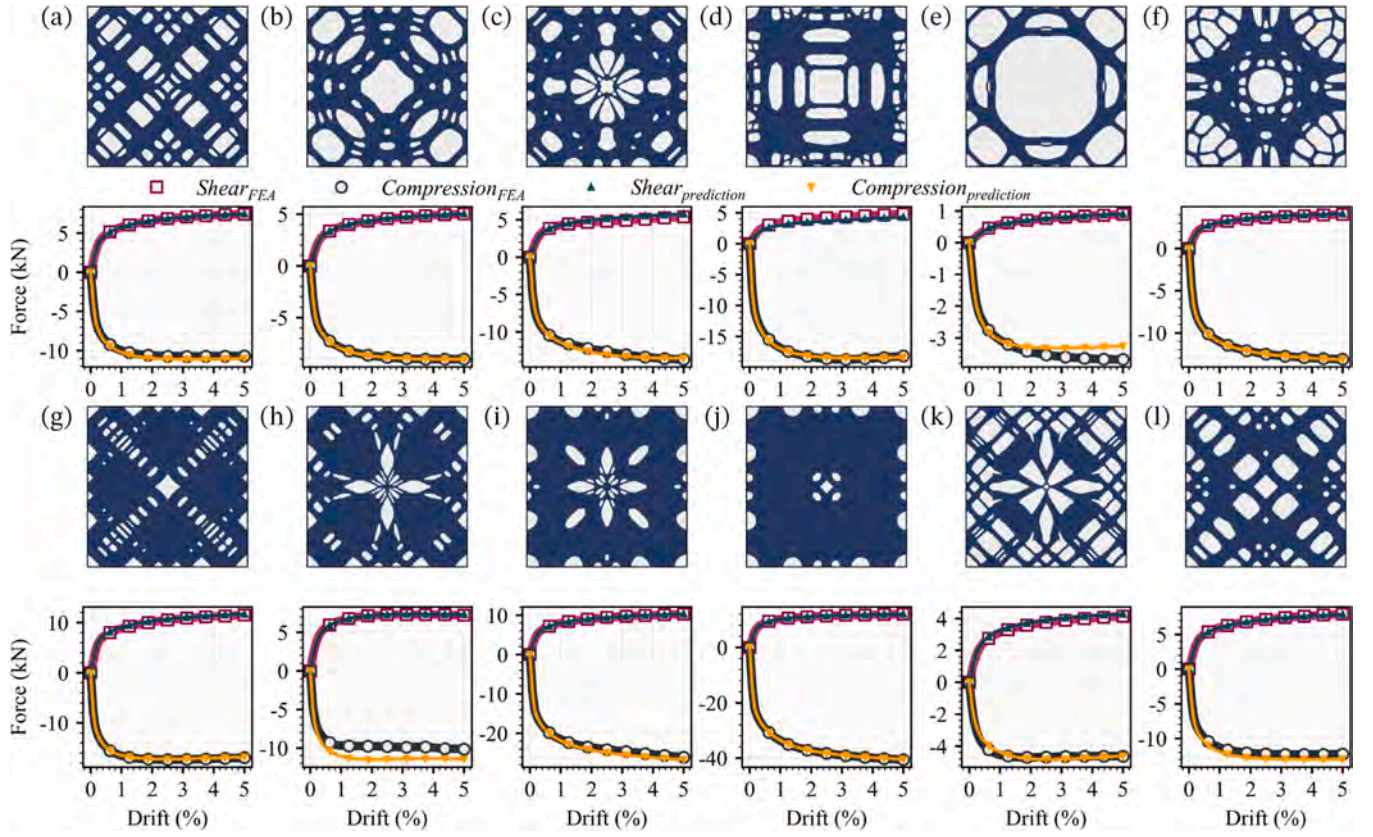


Fig. 13. Prediction results of the ResNeXt-based discriminator.

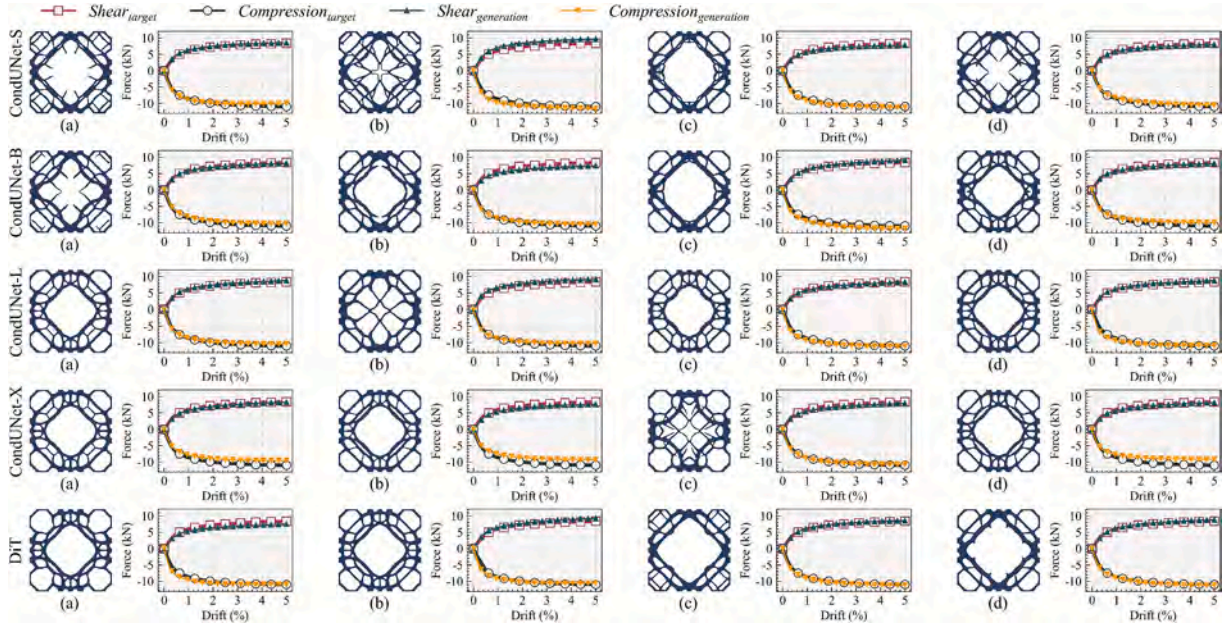


Fig. 14. Generated results of different models for the same target.

performance.

For 1200 sets of mechanical performance targets, generative tests were conducted using five proposed diffusion model frameworks. The FID, CONV, PREC, and SSIM of each sample at different denoising steps were recorded, as illustrated in Fig. 16. The number of denoising steps was set to 1000, identical to the number of noise addition steps, thereby ensuring sufficient convergence of the generated samples. FID quantifies

the closeness between the feature distribution of the generated microstructures and that of the real ones, and it is strongly correlated with human perceptual evaluation of image quality. A good performance typically corresponds to values in the range of 10 to 50, indicating that the proposed generative models can accurately capture the target distribution. Furthermore, the four curves of Impact of denoising steps on quantitative evaluation metrics demonstrate that, under the total of

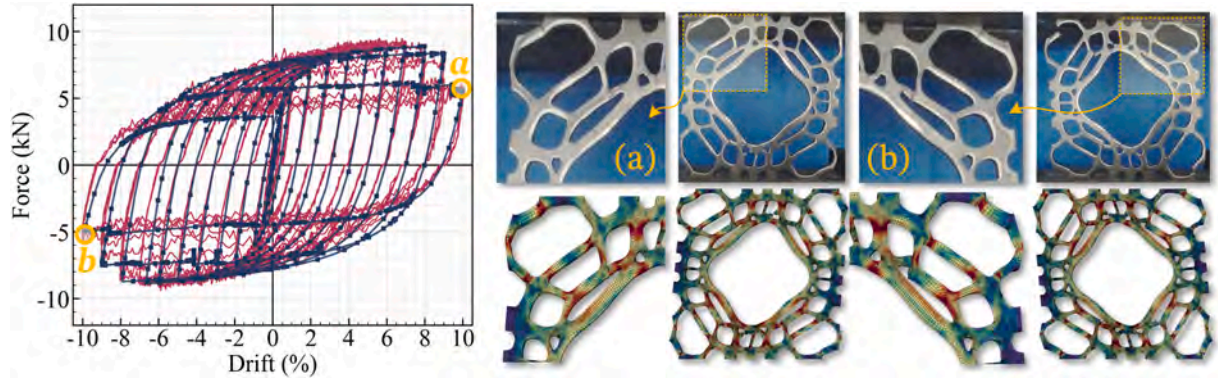


Fig. 15. Validation of generated results.

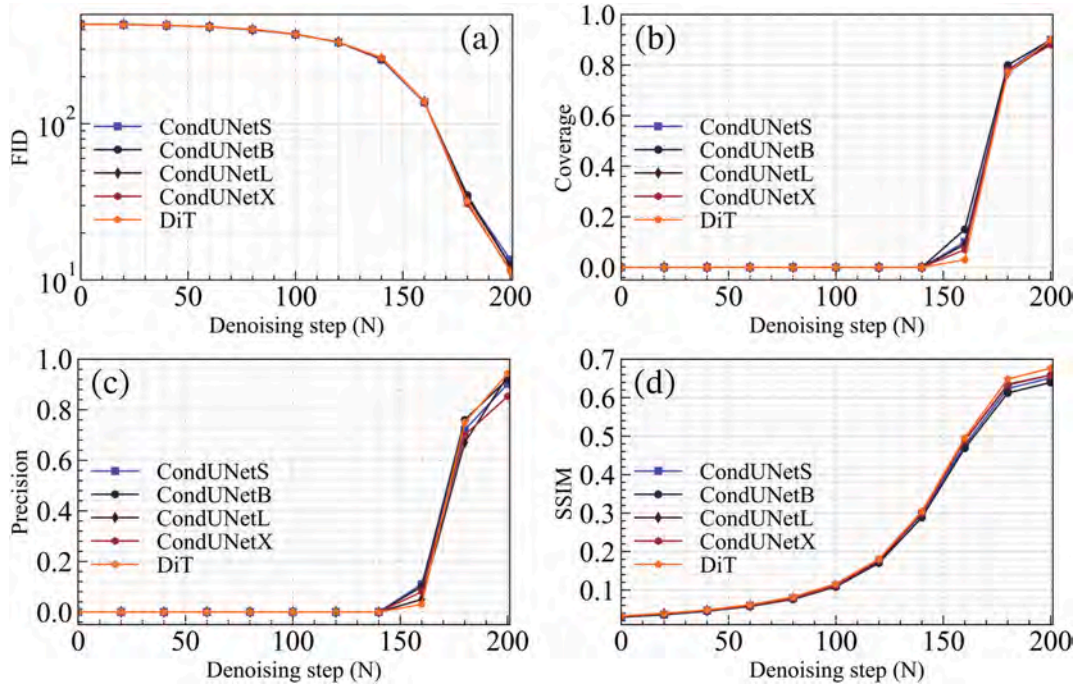


Fig. 16. Impact of denoising steps on quantitative evaluation metrics.

1000 denoising steps, the quality of the generated results improves sharply from approximately the 800th step, and satisfactory outcomes are consistently obtained by the 1000th step.

The quantitative performance of different microstructure generation models in terms of computational complexity and structural quality metrics is summarized in Table 4. For the CondUNet variants, the model size increases from 24.33 M to 99.91 M parameters, accompanied by corresponding growth in FLOPs and MACs. As the scale expands, the FID generally decreases, indicating improved fidelity of the generated microstructures, with CondUNet-X achieving an FID of 11.592. However, trade-offs are observed between COV and PREC across different variants, suggesting that larger models do not necessarily guarantee

balanced performance. In particular, CondUNet-L achieves the highest precision of 0.934 but exhibits relatively lower coverage. By contrast, the DiT model, with 138.09 M parameters, delivers the best overall performance, achieving the lowest FID of 11.367, the highest SSIM of 0.676, and a favorable balance between precision and coverage. These findings highlight the effectiveness of the DiT architecture in capturing the target distribution.

5.3. Generation process

Since the DiT-based damping microstructure generation framework achieved the best performance across all quantitative evaluation

Table 4
Quantitative results of different models.

	Params (M)	FLOPs (G)	MACs (G)	FID	COV	PREC	SSIM
CondUNet-S	24.33	2.91	1.46	13.535	0.869	0.902	0.651
CondUNet-B	39.75	4.31	2.15	12.380	0.862	0.921	0.639
CondUNet-L	84.67	7.06	3.53	12.373	0.850	0.934	0.658
CondUNet-X	99.91	4.43	2.21	11.592	0.886	0.853	0.658
DiT	138.09	5.49	2.75	11.367	0.889	0.943	0.676

metrics, this model was subsequently adopted to generate damping microstructures under different mechanical performance targets. The generation process of the latent space and the corresponding decoded microstructure configurations is illustrated in Fig. 17.

At the initial stage of the denoising process, when only a single denoising step is applied, the latent space essentially corresponds to pure Gaussian random noise. In this case, the decoded microstructures exhibit no recognizable regularity or connectivity. The generated patterns are highly fragmented, with different regions almost entirely disconnected, and the structural boundaries fail to follow the expected continuity, rendering the samples physically unrealistic. As the number of denoising steps increases, the latent representations gradually evolve into more structured forms. For instance, when the denoising steps reach approximately 125, the generated microstructures begin to display discernible regularity. At this stage, the generative model first outlines the boundary regions of the samples and subsequently refines the internal details. This hierarchical generation process reflects the model's ability to capture both global geometric features and local fine-scale characteristics. Ultimately, when the denoising steps reach 200, the model is capable of producing high-quality microstructures with clear connectivity and structural fidelity. At this point, the latent space shows a strong correlation with the final microstructure, where the initial representation may be regarded as a coarse sketch—akin to a reduced version of the final structure but lacking fine details.

To further validate the quality of the generated microstructures, FEA was conducted. The mechanical performance evaluation demonstrates that the generated samples exhibit shear and compressive load-bearing capacities highly consistent with the target reference curves. This similarity extends not only to the elastic stiffness and post-yield stiffness, but also to the overall yield point and corresponding energy dissipation capacity. Such consistency indicates that the generated microstructures are not only visually realistic but also mechanically reliable, thereby confirming the effectiveness of the proposed generative framework.

5.4. Generated microstructures

To further validate the robustness of the proposed microstructure generation framework, eight different mechanical performance targets were randomly specified. For each target, eight independent generations were performed, resulting in a total of 64 microstructures, as illustrated in Fig. 18. In theory, each generated microstructure should be distinct. However, certain samples exhibit partial visual similarity. For example, in Fig. 18(a), samples I and II share an overall structural resemblance, but differ in the gaps located near the four boundary regions. Similarly, in Fig. 18(c), samples I and II appear highly similar, with a SSIM of 0.89, although closer inspection reveals noticeable differences in the curvature along their boundaries.

FEAs under shear and compressive loading were conducted for all 64

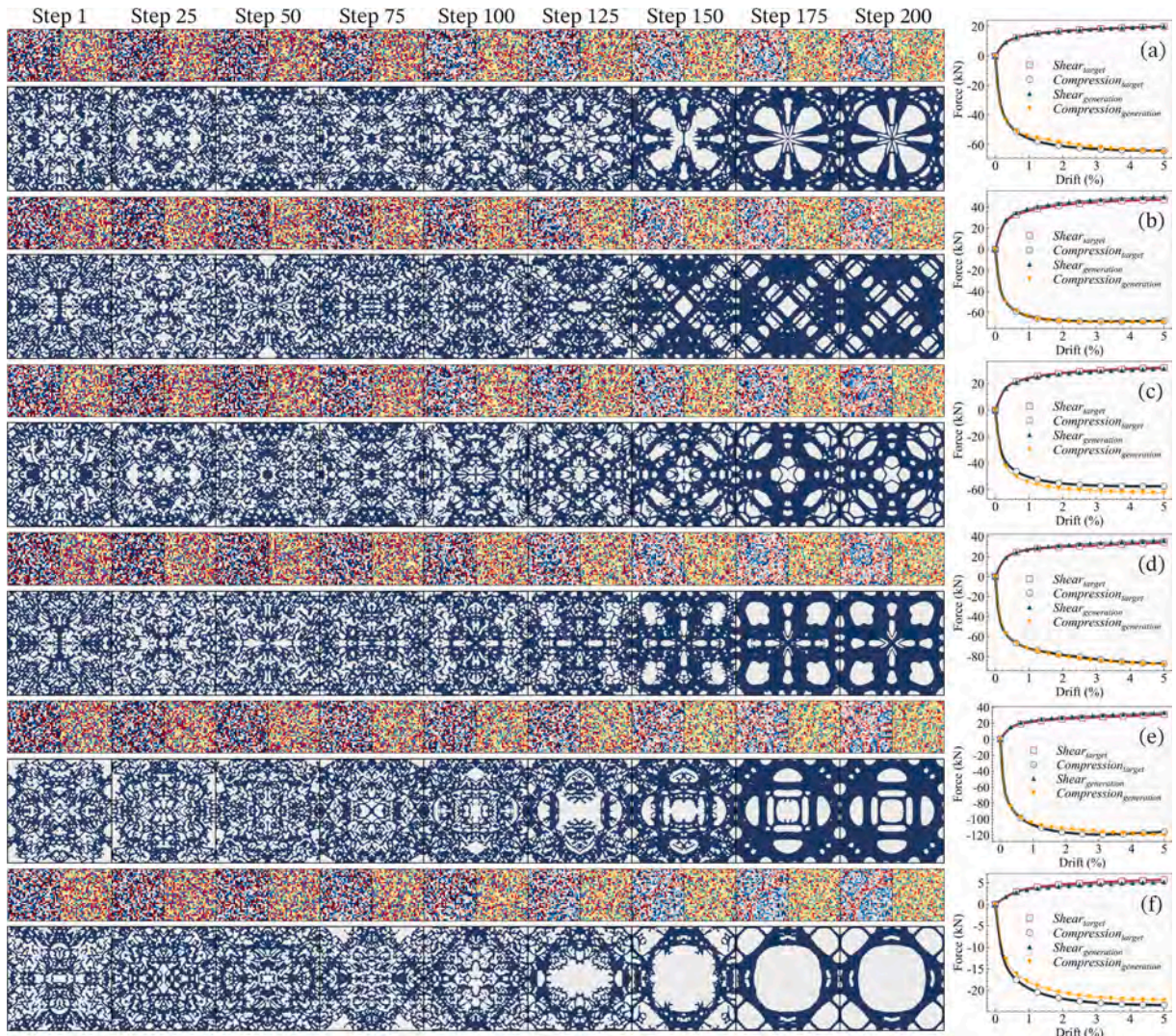


Fig. 17. Damping microstructure generation process.

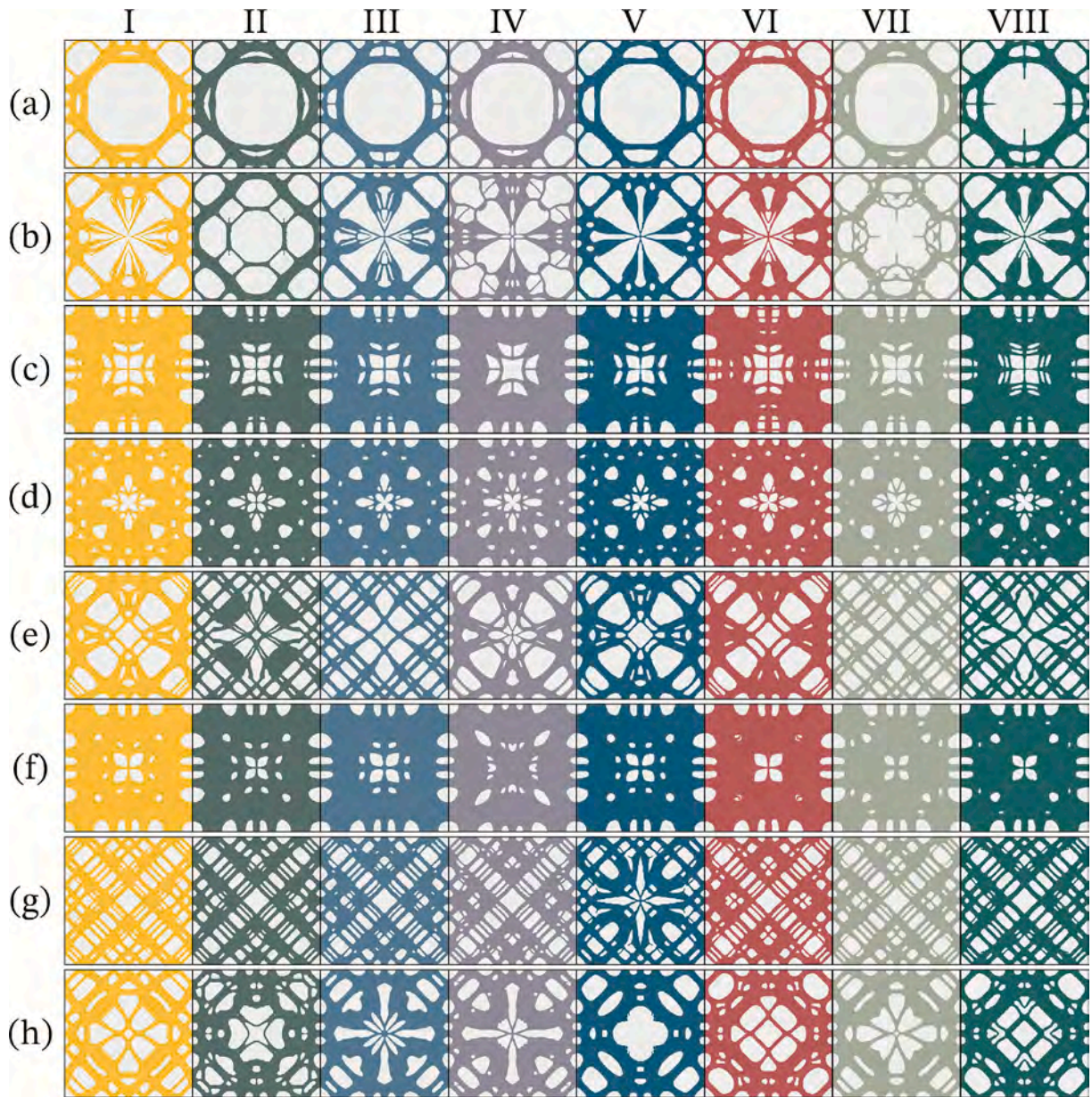


Fig. 18. Generated microstructures.

generated microstructures, and the corresponding mechanical responses are presented in Fig. 19. The colors of the performance curves are consistent with the labels shown in Fig. 18. In general, the proposed generation framework exhibits strong capability in satisfying diverse nonlinear mechanical performance requirements, as most generated microstructures closely follow the target curves. To further verify whether the generated results originate directly from the training dataset, we conducted a similarity analysis between the 64 generated microstructures shown in Fig. 18 and all samples in the dataset using the SSIM. The results show that the maximum SSIM values are far below 1.0, indicating that even in the most similar cases, the generated structures maintain clear geometric differences from any training sample. This quantitative evidence further supports our conclusion that the model generates novel designs rather than simply replicating existing data.

However, deviations are also observed. For example, the yellow curve in Fig. 19(b), corresponding to sample I in Fig. 18(b), shows a large discrepancy from the target. Although sample I has high structural similarity with samples II and VI (SSIM values of 0.68 and 0.66, respectively), its load-bearing capacity is significantly lower. The

difference arises because the internal structure of sample I is not fully connected. Such a seemingly minor defect results in substantial performance differences, with compressive strength reductions of 30.38 % and 19.45 % compared with samples II and VI, respectively. These results highlight the sensitivity of the generation process to local structural imperfections and justify the incorporation of a ResNeXt-based discriminator to constrain the generation and improve robustness. On the other hand, when the target requires relatively higher load-bearing capacity, the framework performs more reliably. For instance, in Figs. 19(c) and (d), where compressive strength exceeds shear strength, and in Figs. 19(e) and (g), where the two are comparable, the generated microstructures match the targets well. Overall, the framework effectively achieves the one-to-many mapping from nonlinear mechanical performance objectives to feasible microstructural configurations.

In addition to the qualitative comparison of target and generated curves, a quantitative error analysis was performed for all 64 generated microstructures, as summarized in Table 5. The metrics include RMSE, MPAE, yield force error, maximum force error, stiffness error, and energy dissipation error, offering a comprehensive evaluation of

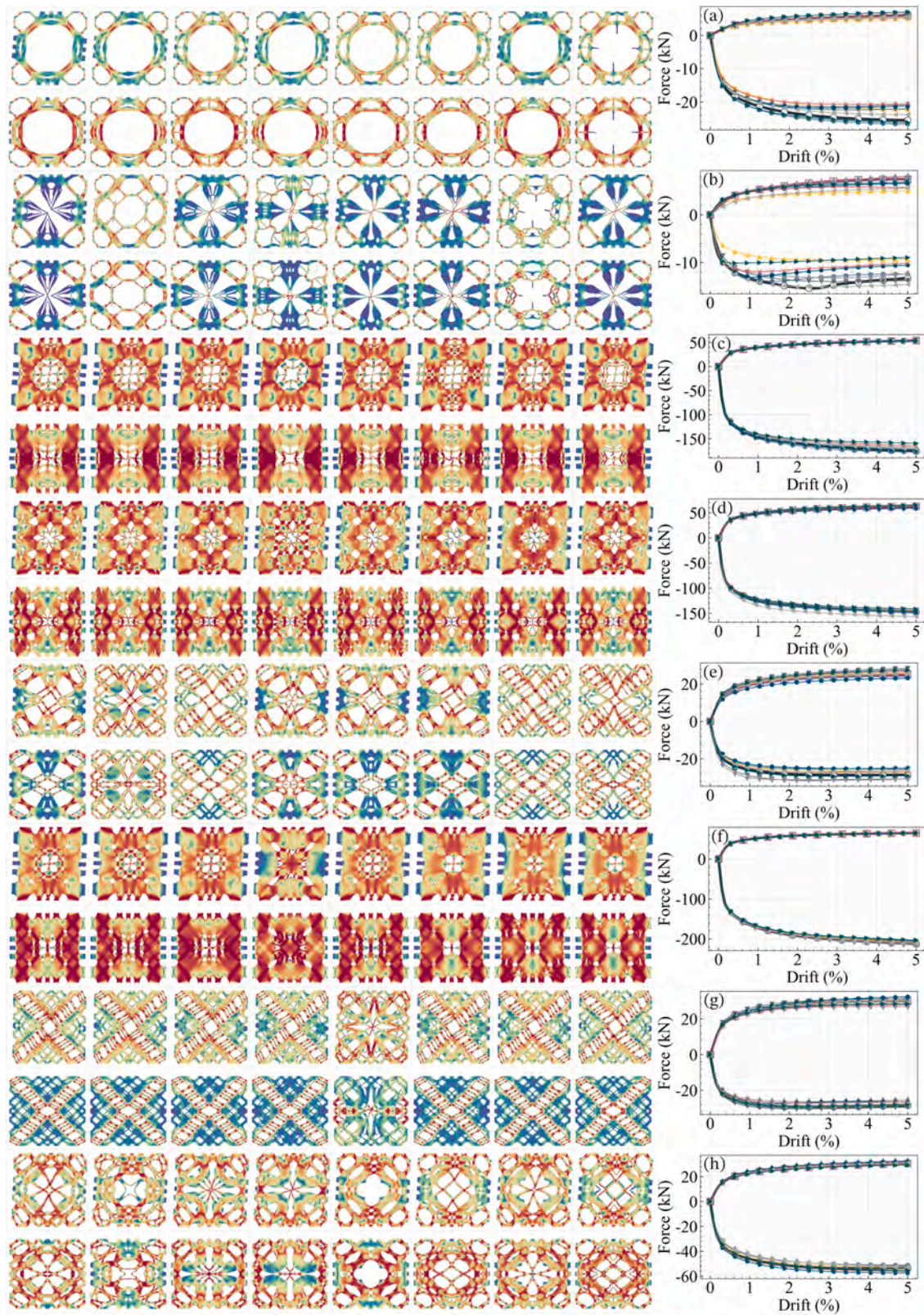


Fig. 19. Finite element analysis verification.

mechanical fidelity. Overall, RMSE values mostly fall within the range of 1 to 3 kN, indicating close tracking of target curves throughout the loading history. MPAE values are generally below 5 % for the majority of groups, with slightly higher deviations observed in cases requiring large stiffness change. Yield and maximum force errors typically remain

within ± 15 %, confirming that strength levels are reproduced with reasonable accuracy. Stiffness errors are low in most groups, though certain high-stiffness targets exhibit larger deviations, reflecting sensitivity to local geometric variations. Energy dissipation errors mostly stay under 10 %, validating the preservation of damping capacity. These

Table 5

Error statistics of mechanical performance for generated microstructures.

Figure	RMSE (kN)	MPAE (%)	Error _{yield force} (%)	Error _{max force} (%)	Error _{stiffness} (%)	Error _{energy} (%)
a	1.31	4.50	10.57	10.62	6.49	9.13
b	1.63	7.41	14.79	15.84	28.84	14.17
c	2.82	1.14	2.28	2.71	2.15	2.28
d	2.57	1.45	3.08	3.16	2.91	2.89
e	1.74	3.50	6.68	6.79	10.45	6.86
f	2.18	0.84	1.46	1.36	3.55	1.60
g	1.13	1.98	4.16	4.25	4.37	3.95
h	1.44	1.78	3.28	3.32	6.12	3.34

quantitative results, consistent with the visual observations, demonstrate that the proposed framework achieves high fidelity in matching the desired mechanical performance while maintaining structural diversity.

6. Conclusion

This study proposes an integrated generative framework for the efficient and high-fidelity inverse design of damping microstructures in EDSW for seismic applications. The framework establishes a seamless pipeline from pixel-based microstructure datasets to latent space modeling, three-dimensional reconstruction, industrial fabrication, and FEA validation. The main conclusions of this study are as follows:

1. By employing feature compression through a variational autoencoder-based TopoFormer and dataset simplification via clustering, the generation complexity is reduced by more than 90 % while maintaining high reconstruction fidelity. This facilitates the subsequent learning of the mapping between mechanical performance and geometry using the conditional latent DiT.
2. The conditional DiT, trained with full nonlinear shear and compressive performance curves as conditional inputs, achieves accurate one-to-many mappings from target hysteretic responses to manufacturable geometries. It obtains an FID of 11.367 and an SSIM of 0.676, while maintaining balanced accuracy and coverage, significantly outperforming the attention-based CondUNet baseline models.
3. Experimental validation using laser-cut specimens under low-cycle reciprocating loading confirmed a strong consistency among the measured, theoretical, and generated hysteresis curves. This demonstrates that the generated microstructures satisfy both geometric and mechanical performance requirements and verifies their practical feasibility for application in the novel seismic energy-dissipation system EDSWs.

While this study focuses on the comparison between CondUNet and Transformer-based DiT architectures in the context of damping microstructure generation, it is worth noting that other generative paradigms have also achieved notable results in related domains. GAN-based models have demonstrated strong capability in producing visually realistic microstructures, but often face challenges in mode collapse and in capturing complex nonlinear mechanical targets without additional constraints. Physics-informed generative models, on the other hand, embed governing equations or surrogate physical models into the generation process, potentially improving mechanical consistency but sometimes limiting geometric diversity. Compared with these approaches, our conditional latent DiT framework emphasizes the direct mapping between full nonlinear performance curves and manufacturable geometries, achieving balanced fidelity and diversity without requiring explicit physical solvers during generation. Nevertheless, a limitation of the present work is that the evaluation is restricted to pixel-based datasets and VAE-encoded latent spaces. Future work could integrate physics-informed constraints or hybrid generative architectures to further enhance mechanical consistency and extend

applicability to more complex, multi-scale structural systems.

CRedit authorship contribution statement

Tianyang Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Weizhi Xu:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Shuguang Wang:** Writing – review & editing, Supervision, Resources, Methodology. **Dongsheng Du:** Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The related codes are available at <https://github.com/AshenOneme/DiT-Based-Microstructures-Design>.

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